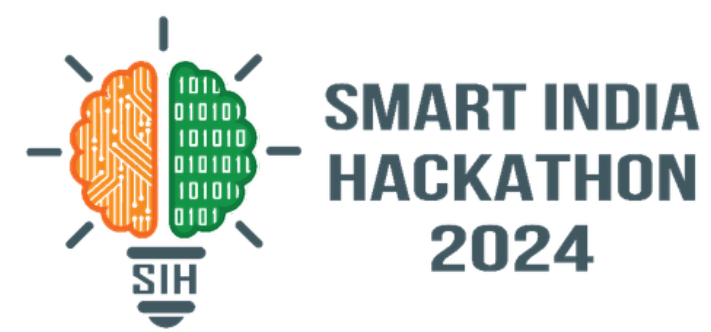


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH 26164**
- **Problem Statement Title- Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)**
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID-**
- **Team Name- Mudrita**



Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)



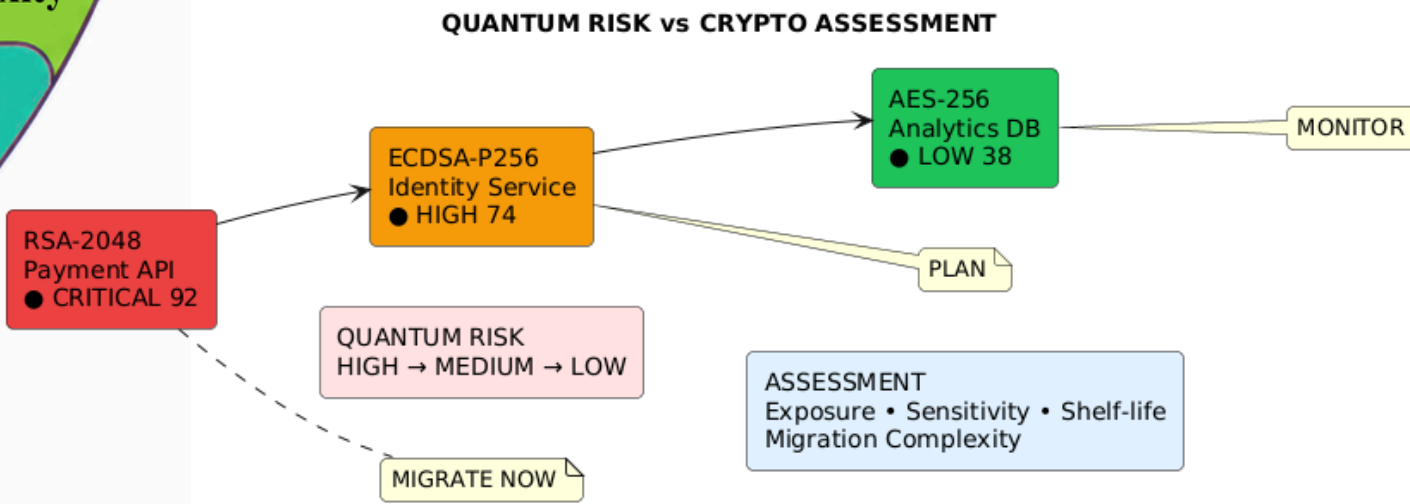
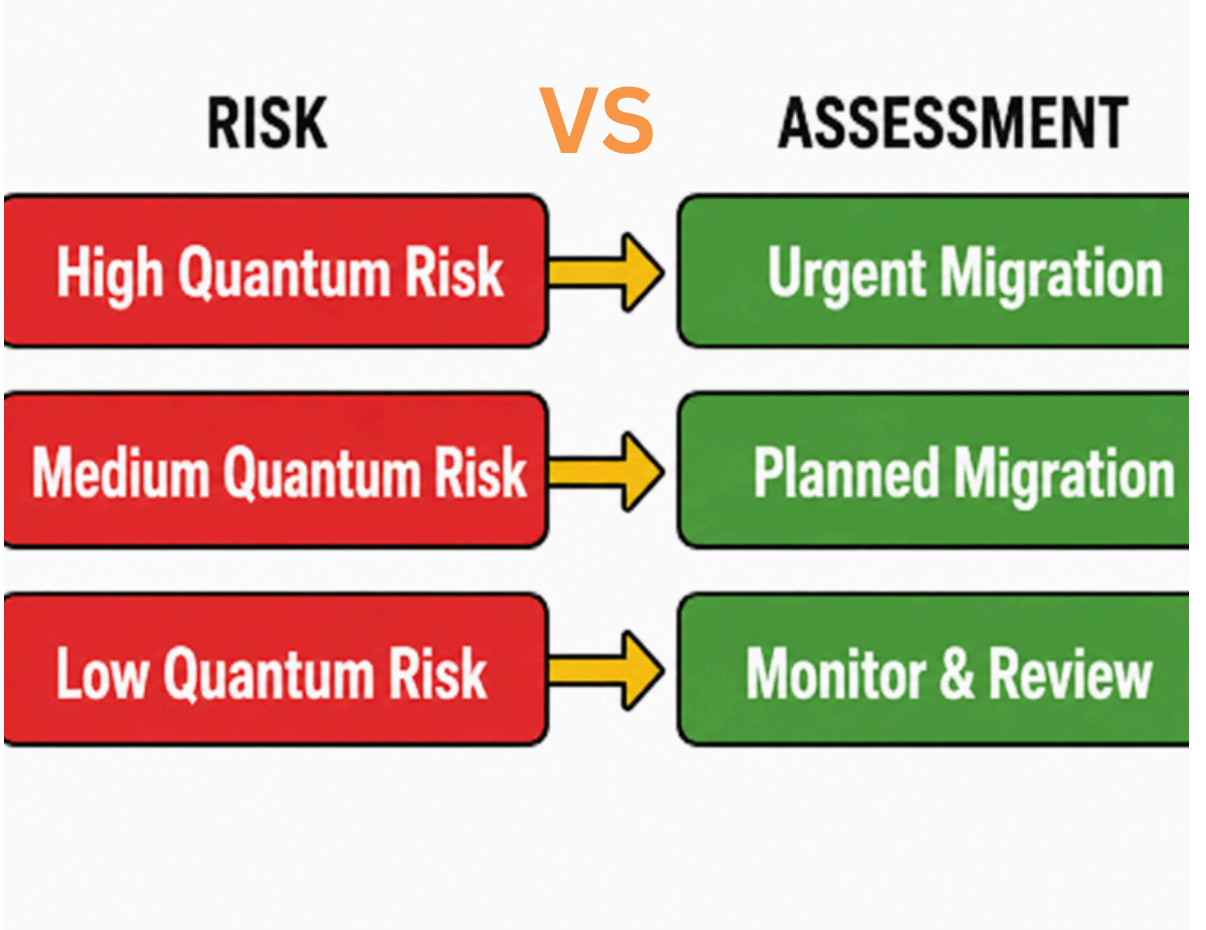
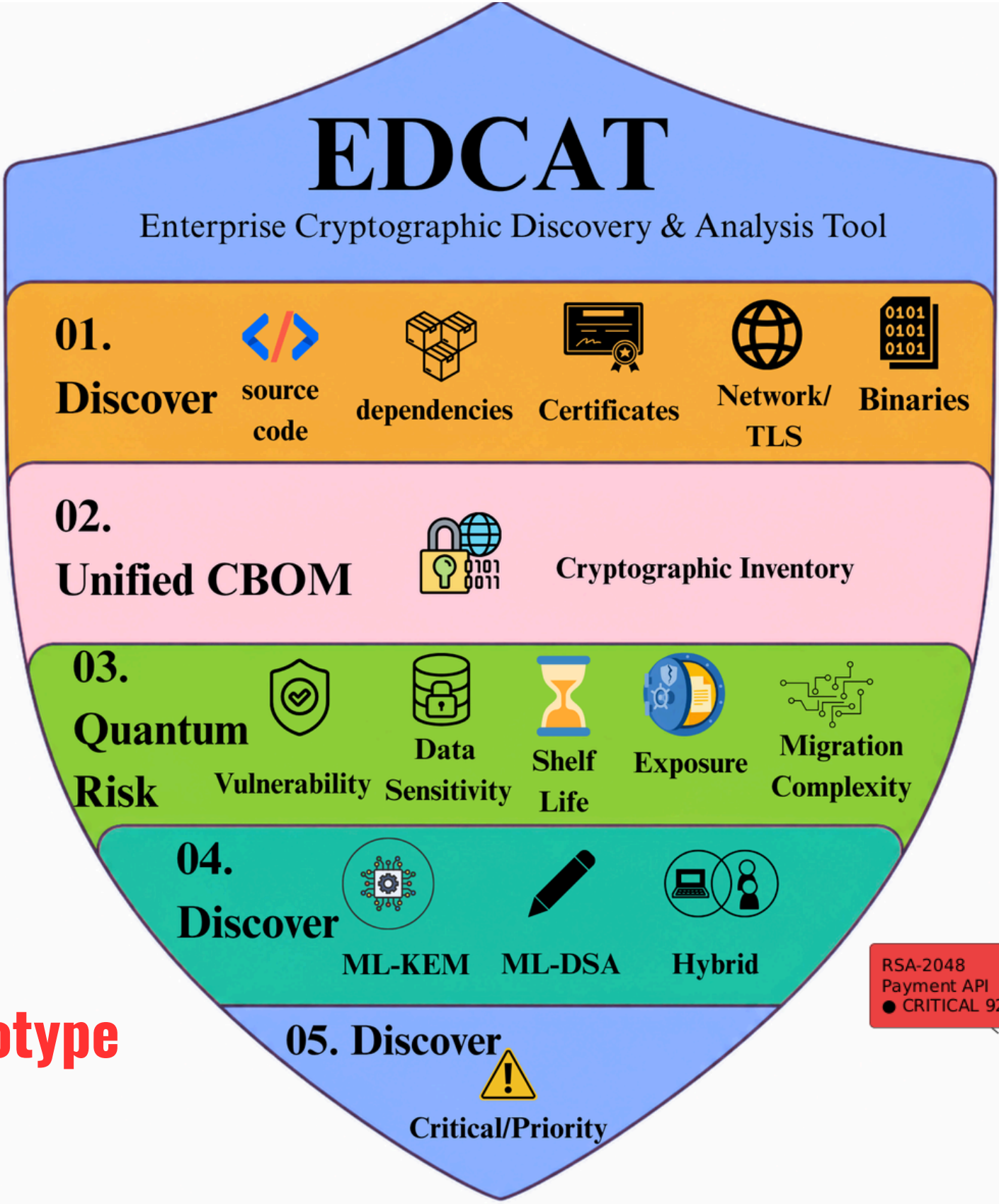
The problem: Crypto is scattered across enterprise systems.
No unified visibility or migration priority.

Why It Matters?: Quantum-vulnerable crypto threatens long-lived sensitive data.
Limited migration time increases future exposure.

The Solution : ECDAT discovers and analyzes enterprise cryptography.
It scores risk, recommends PQC, and prioritizes migration

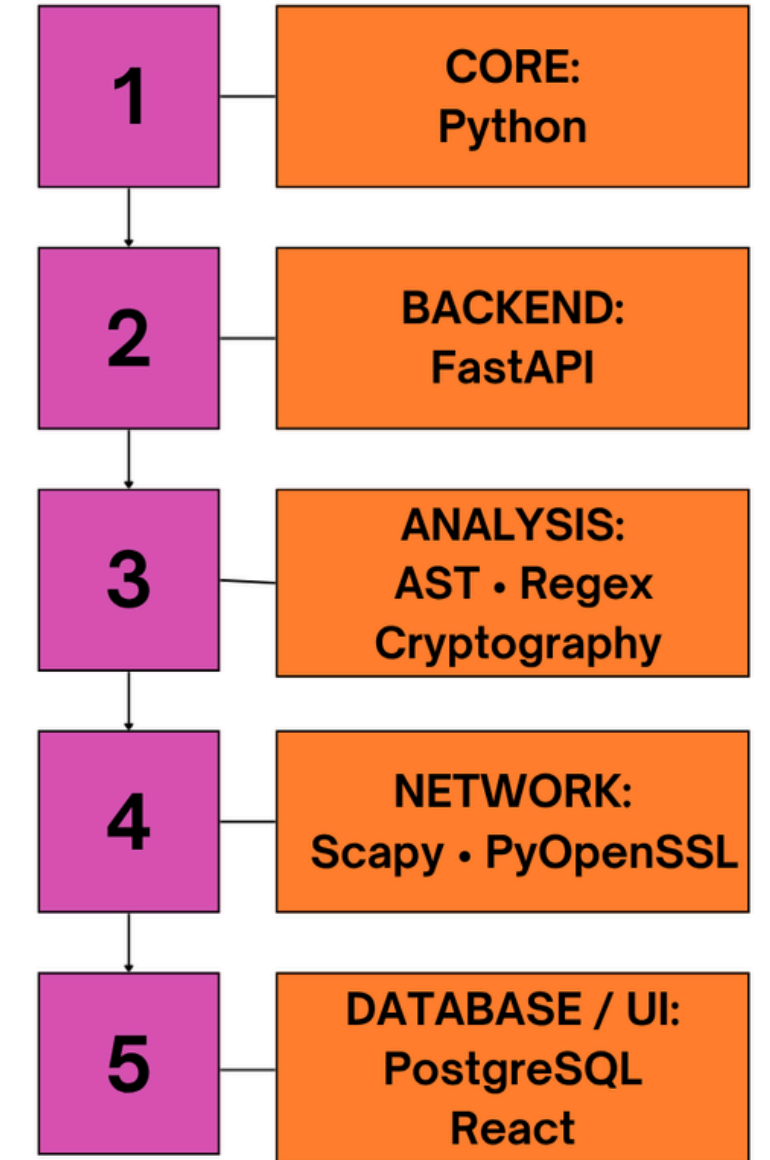
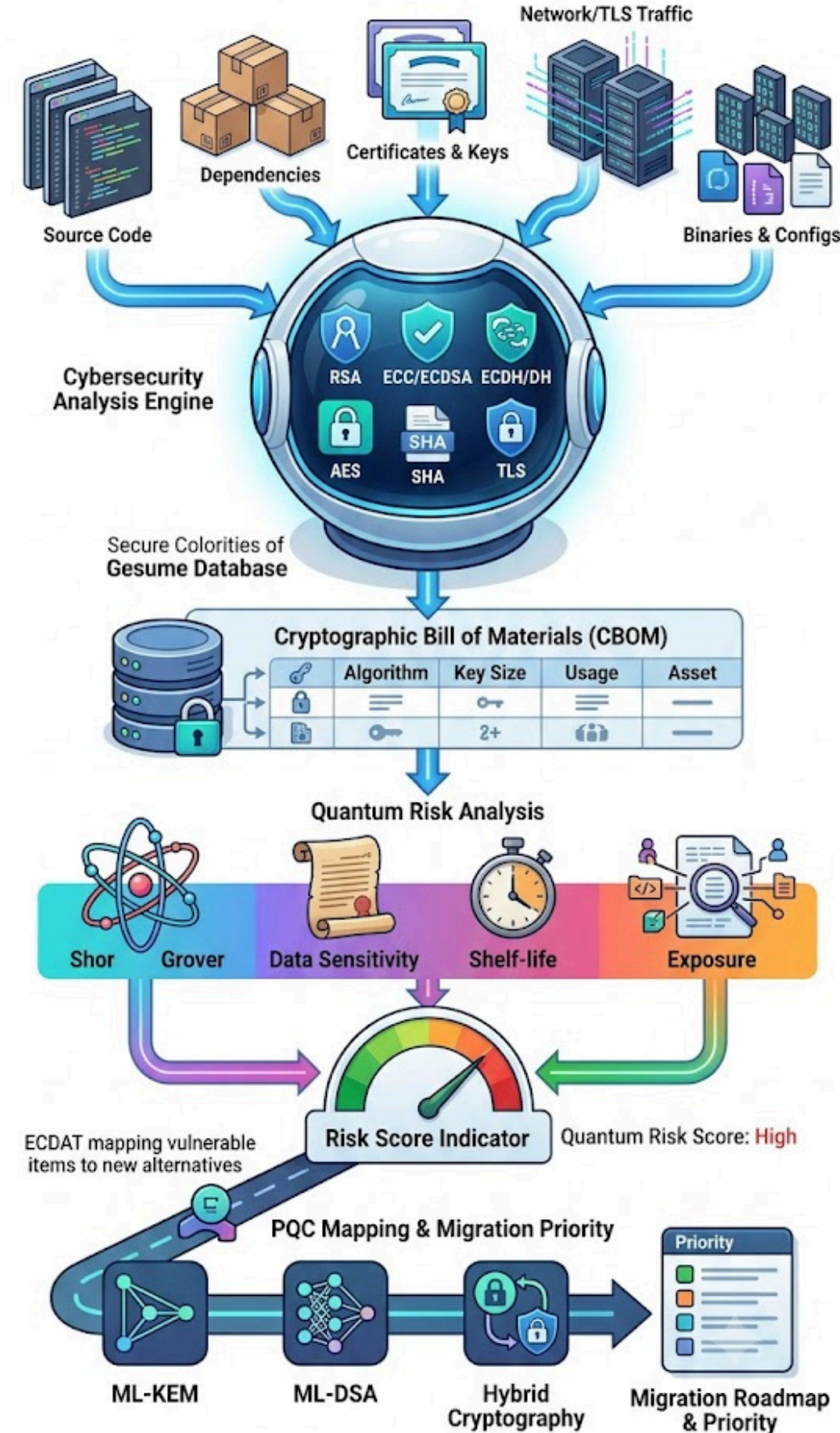
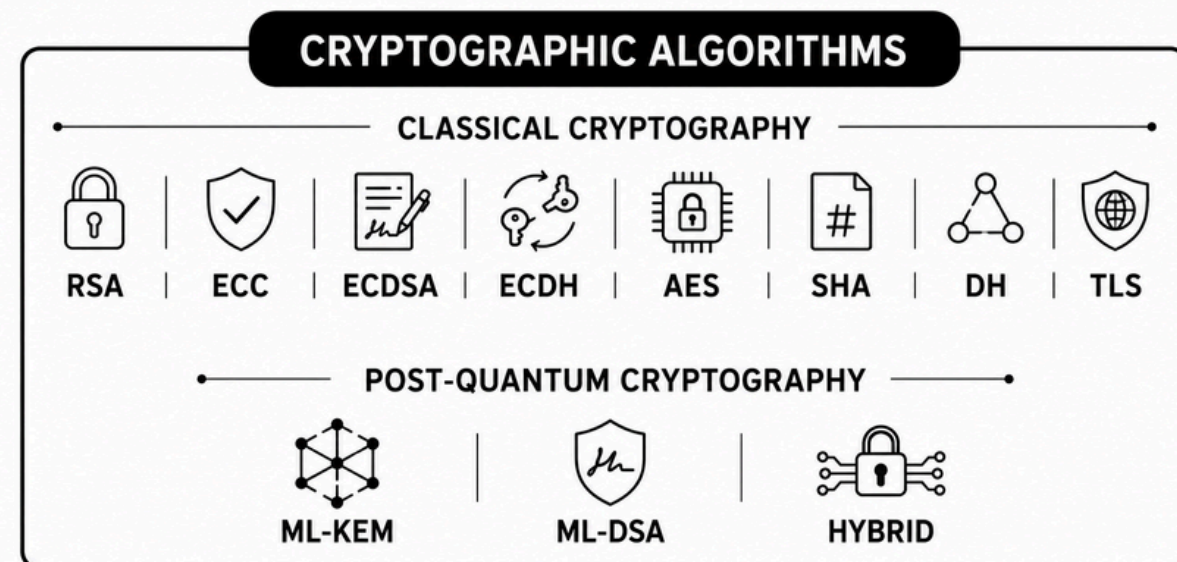
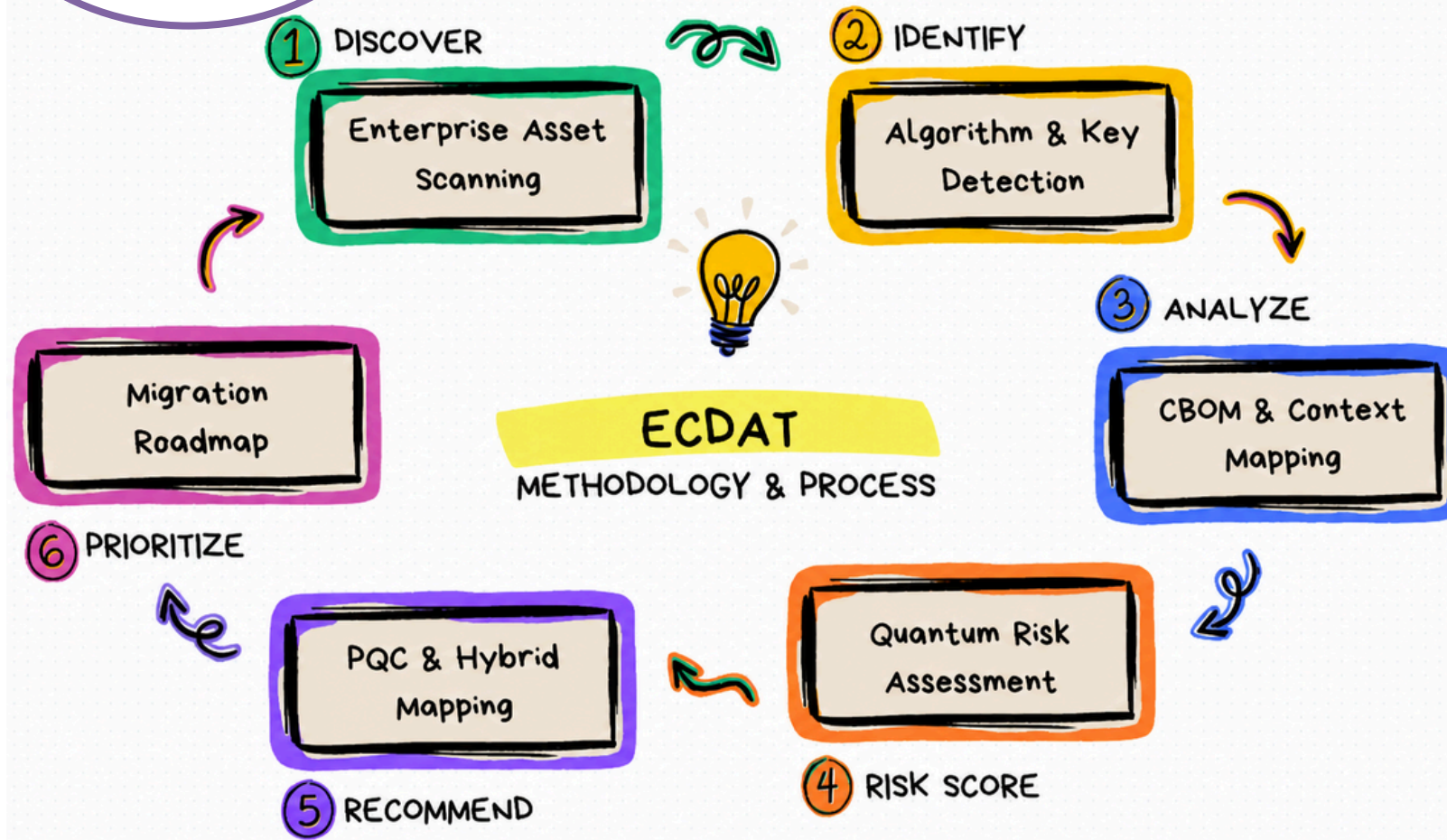
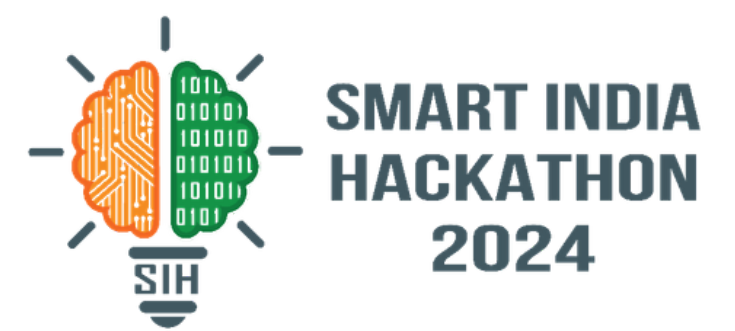


Prototype



Mudrita

TECHNICAL APPROACH



Detailed
Report

[GITHUB
REPORT](#)

FEASIBILITY AND VIABILITY

TECHNICAL FEASIBILITY

Existing Technology

Built using mature, open-source technologies such as Python, FastAPI and React.

Open-Source Analysis

AST, Regex, OpenSSL and Scapy enable practical code, cryptographic and network analysis.

Modular Architecture

Independent scanning and analysis modules allow easy extension for new asset types and algorithms.

Standardized Output

CBOM provides a structured format for consolidating cryptographic findings across diverse environments.

OPERATIONAL VIABILITY

Enterprise-Wide Discovery

Scans source code, binaries, libraries, containers and network-facing assets from a unified platform.

Actionable Risk Assessment

Combines cryptographic usage with data sensitivity, shelf-life, exposure and business criticality.

Migration Decision Support

Provides prioritized PQC or hybrid recommendations based on risk, impact and migration complexity.

Standardized Reporting

Generates CBOM-based reports that security, development and compliance teams can use for remediation planning.



BUSINESS POTENTIAL

Enterprise Adoption

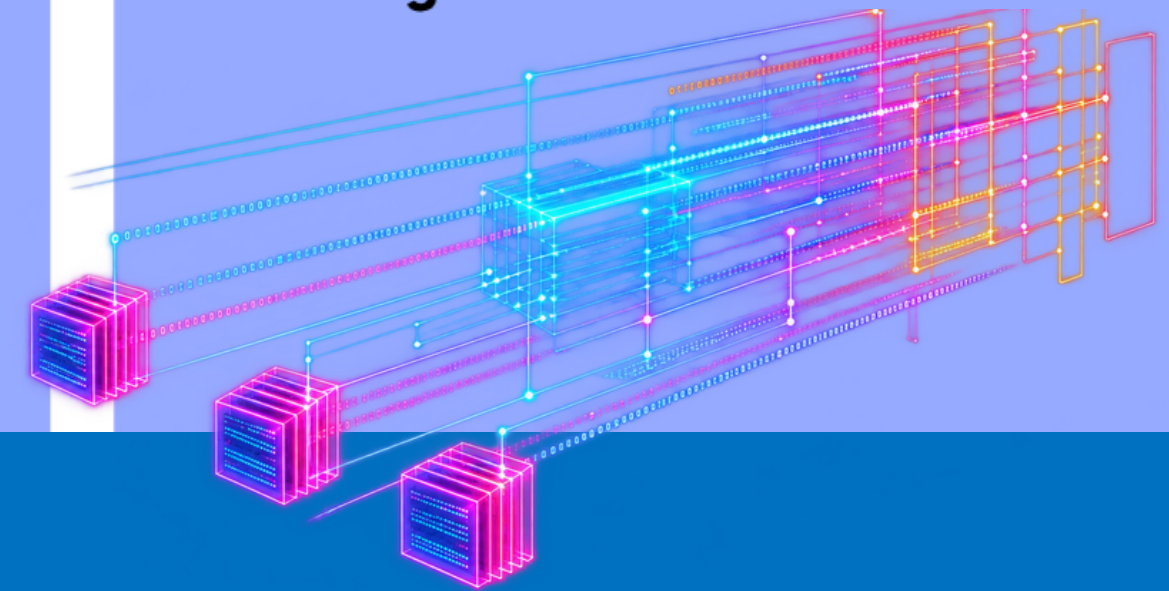
Scale across applications, infrastructure and cloud environments.

PQC Migration Readiness

Enable risk-based, cost-aware migration to PQC and hybrid solutions.

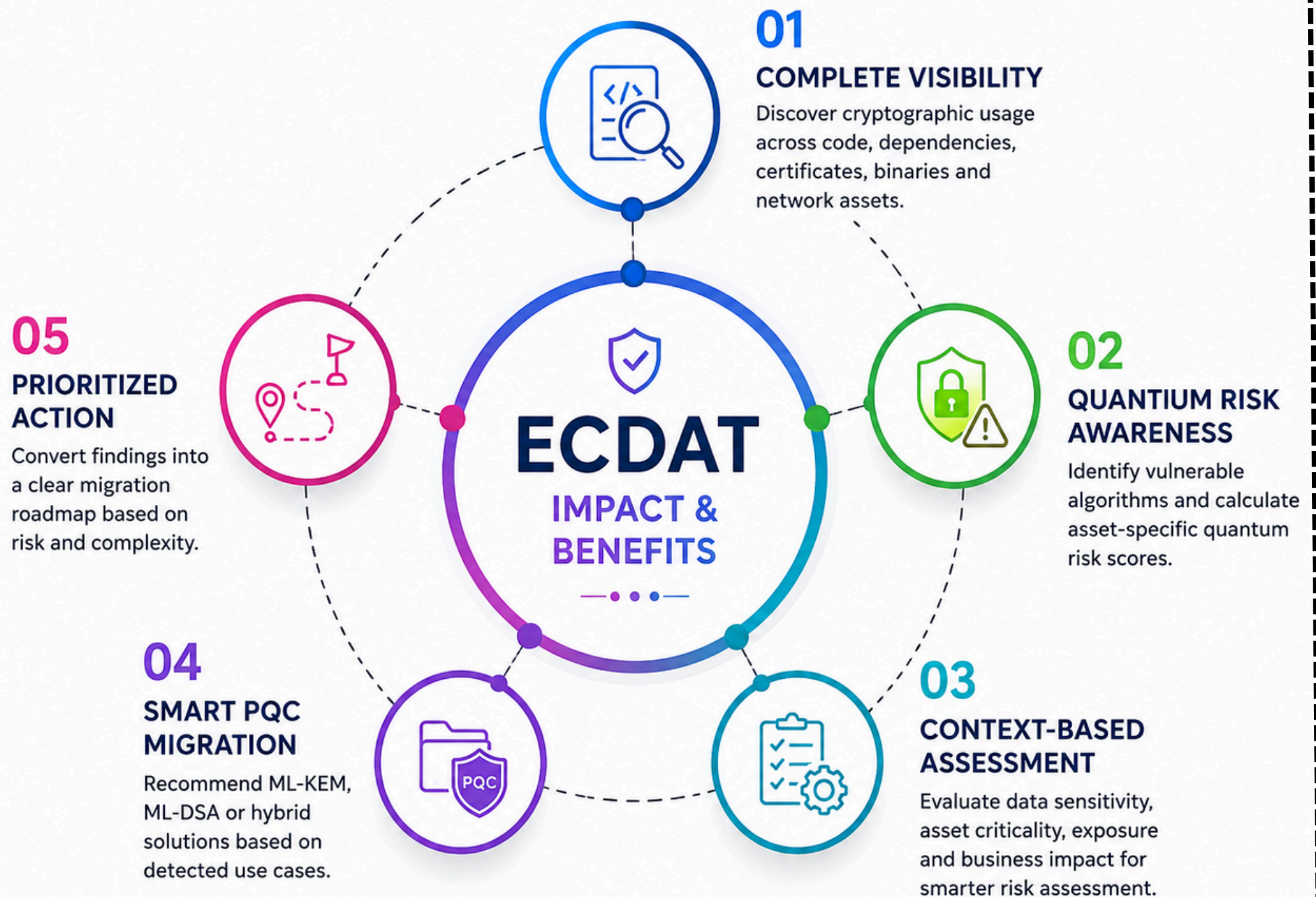
Continuous Crypto Governance

Evolve toward ongoing CBOM management and cryptographic risk monitoring.

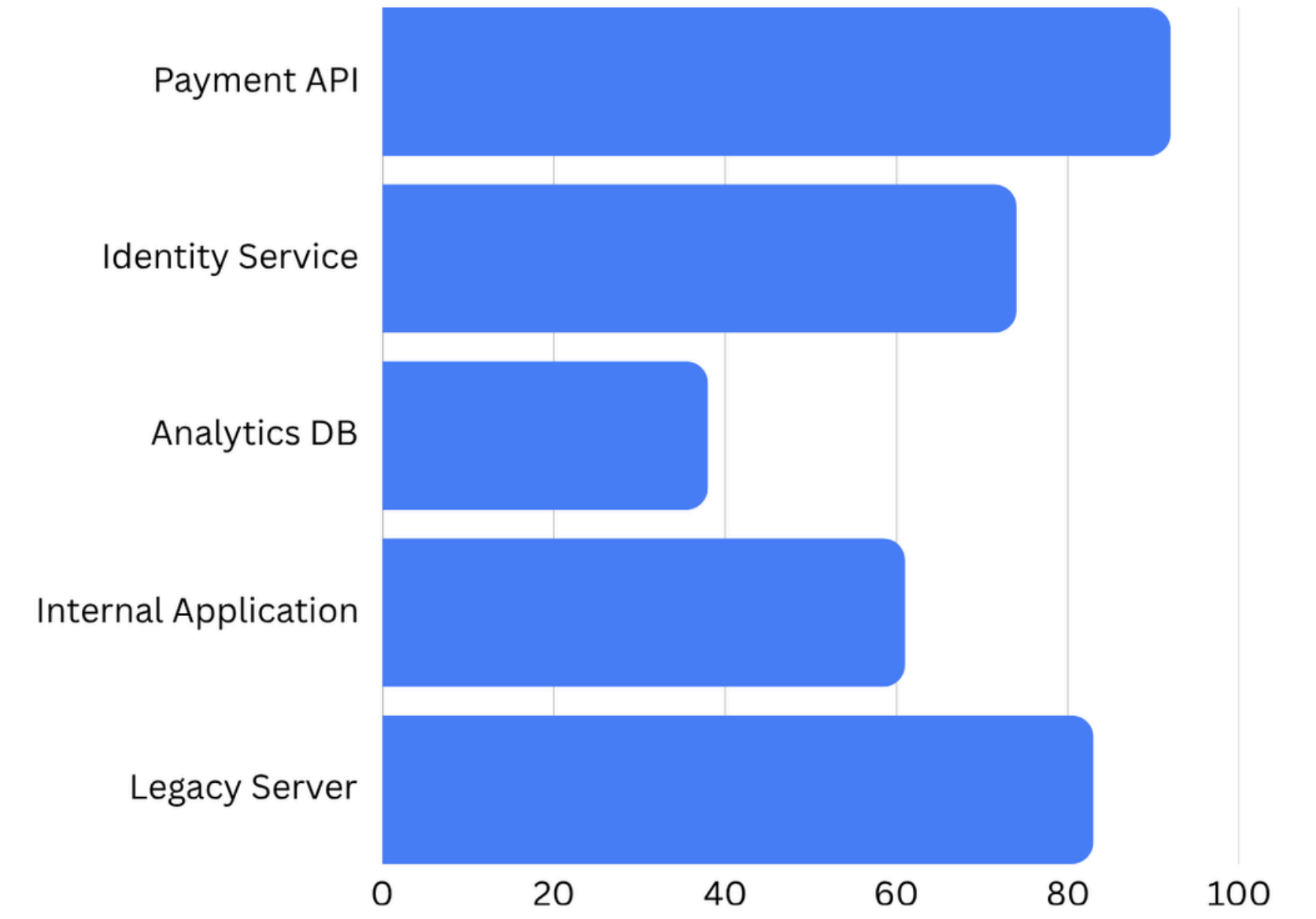


Mudrita

IMPACT AND BENEFITS



QUANTUM RISK-BASED PRIORITIZATION

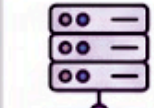


PAYMENT API



RSA-2048 **92 • CRITICAL**

LEGACY SERVER



Legacy Crypto **83 • HIGH**

IDENTITY SERVICE



ECDSA-P256 **74 • HIGH**

RESEARCH AND REFERENCES

CRYPTOGRAPHIC INVENTORIES

Schmitt et al.

Criteria and tooling for discovering cryptographic assets across software, networks, keys and infrastructure.

QUANTUM THREAT TIMELINE

Mosca & Piani

Estimates the potential quantum threat timeline and introduces the relationship between data lifetime and migration time.

MIGRATION TO PQC

NIST

Covers quantum threats, HNDL, risk management, PQC algorithms, performance and migration strategies.

TRANSITIONING ORGANIZATIONS TO PQC

Joseph et al. — Nature

Focuses on organizational requirements and strategies for transitioning to post-quantum cryptography.

ENTERPRISE CRYPTO-AGILITY

ELCA

Addresses enterprise-scale cryptographic configuration, agility and migration across large

NIST PQC STANDARDS

FIPS 203 / FIPS 204

Defines standardized post-quantum algorithms ML-KEM for key establishment and ML-DSA for digital signatures.

CRYPTOGRAPHIC BILL OF MATERIALS

CycloneDX CBOM

Provides a structured approach for documenting cryptographic components, algorithms and

CRYPTOGRAPHIC AGILITY & QUANTUM READINESS

Quantum Readiness Research

Continuous monitoring, lifecycle management & adaptation to evolving

COMPLETE RESEARCH & REFERENCES →

View all papers, standards and sources

ENTERPRISE ASSETS

Code • Network • Infrastructure

DISCOVER

Crypto Detection

CBOM

Unified Inventory

QUANTUM RISK

0-100 Risk Score

PQC

ML-KEM • ML-DSA

PRIORITIZE

Migration Roadmap